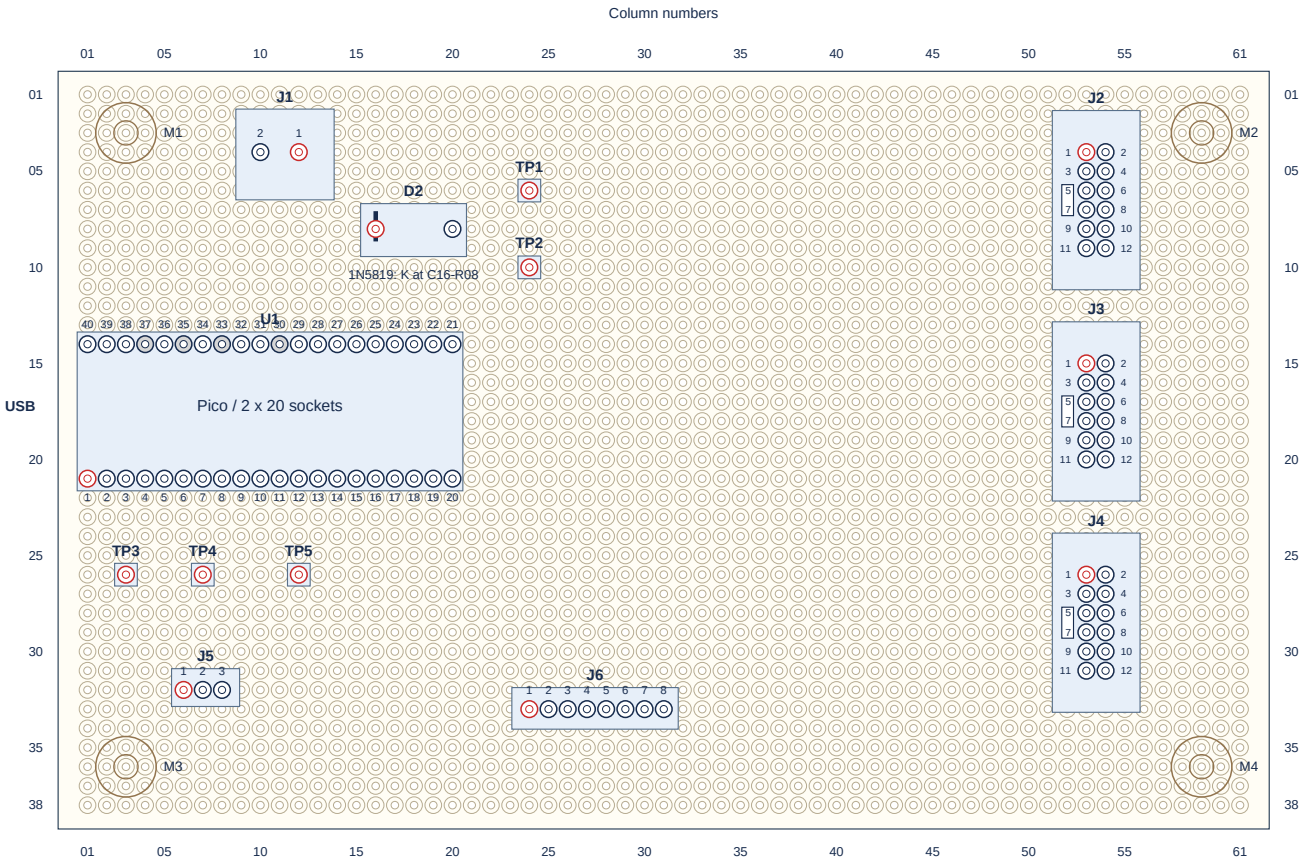
Pico baseboard: component placement

Component side; uncoated face toward you; USB to the left. Coordinates are C(column)-R(row).



Red pad outlines identify pin 1. Grey pads are not externally wired.

Match socket spacing, IDC key orientation and terminal body to the actual purchased parts before soldering.

ASSEMBLY

U1
Pico on 2 x 20 female sockets
Lower socket
C01..C20, R21: pins 1..20
Upper socket
C01..C20, R14: pins 40..21
J1
5V input: 1 positive, 2 GND
D2
1N5819; band/cathode = pin 1
J2 / A
GP0..GP5
J3 / B
GP6..GP11
J4 / C
GP12..GP17
J5
UART: 1 TX, 2 RX, 3 GND
J6
Expansion: see pin-map.csv
TP1 / TP2
VSYS / USB VBUS
TP3 / TP4 / TP5
GP0 / GP1 / GND

J1: enlarge ONLY C12-R04 and

C10-R04 to 1.3 mm for reference terminal.

M1..M4: optional 3.2-mm mounting holes.

Keep a 4-mm radius clear; inspect first.

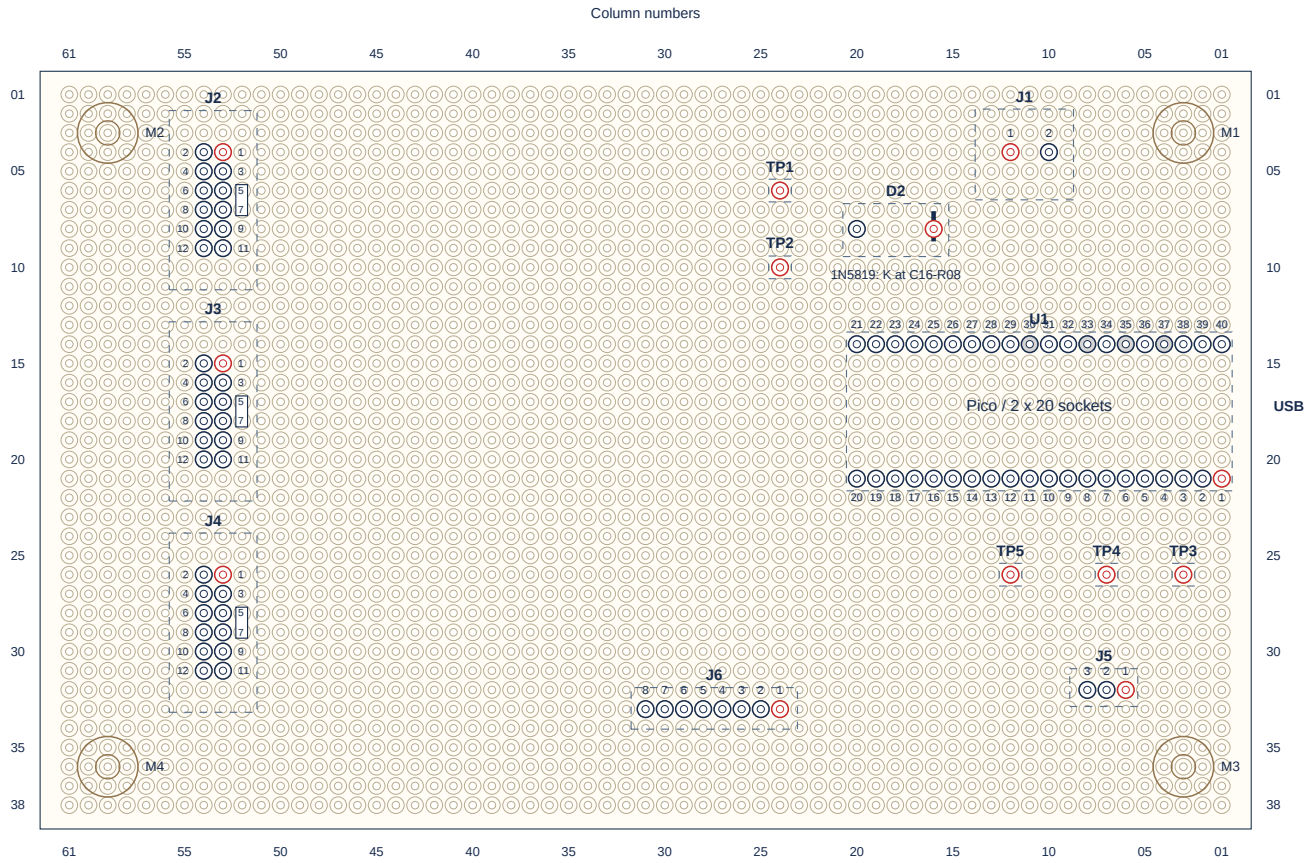
No servo power or shunt on this board.

Print at 100% / Actual size. Disable Fit to page. Check the scale and three distant holes before use.

50 mm scale check

Pico baseboard: solder-side reference

MIRRORED horizontally from component view. USB is now on the right. Hole IDs do not change.



This is the same physical hole map, not a second component placement.

VIEW ORIENTATION

Flip the board left-to-right,

like turning a page.

Rows still run from top to bottom.

Columns run 61 at left to 01 at right.

Ghosted bodies are on the other side.

All wires are insulated.

Crossings are NOT electrical junctions.

Join only the listed endpoints.

U1 socket numbering is deliberate:

physical pins 40..21 on upper row,

pins 1..20 on lower row.

Fit and solder sockets without Pico.

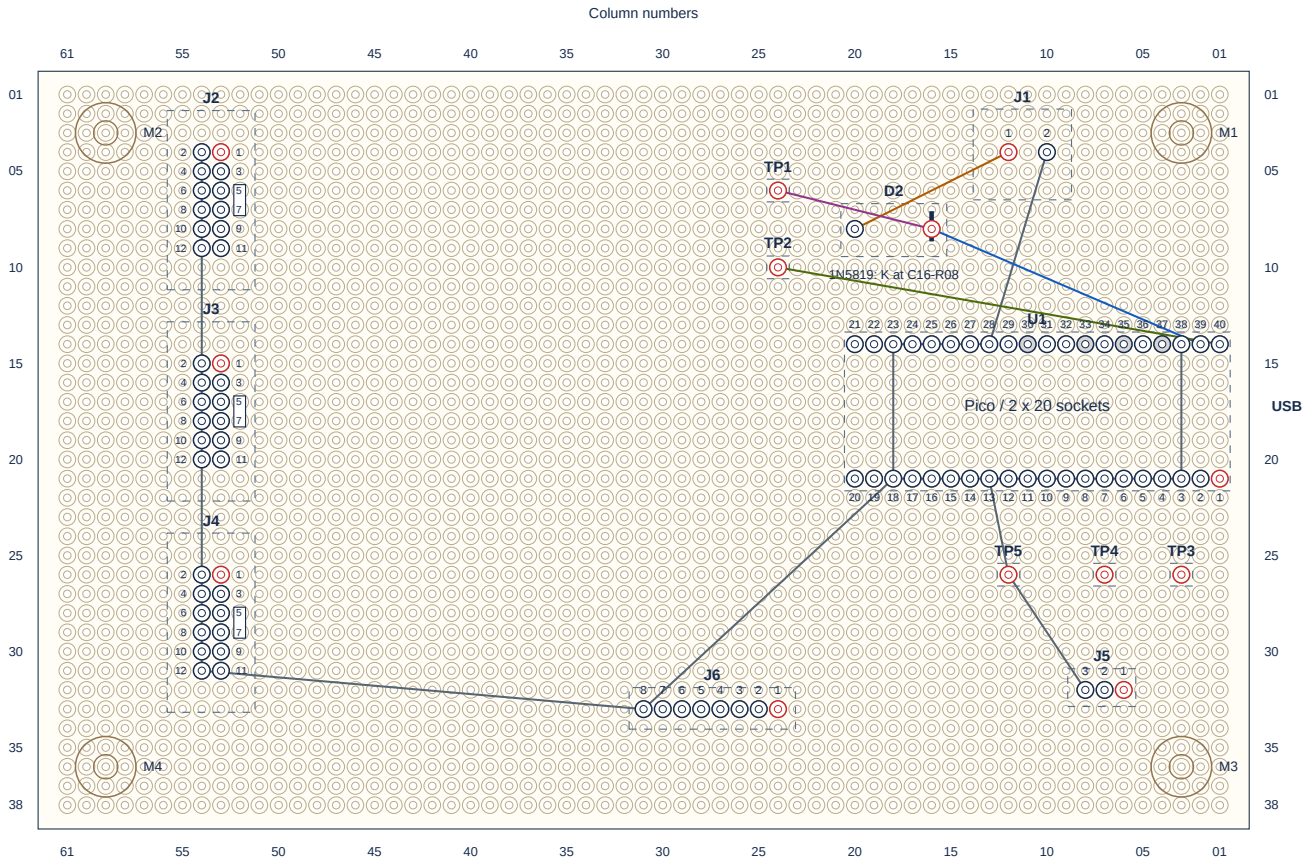
Check wiring before inserting Pico.

Print at 100% / Actual size. Disable Fit to page. Check the scale and three distant holes before use.

50 mm scale check

Solder-side wiring: Ground and Pico supply

Mirrored view. Lines show endpoint connections; choose a practical insulated-wire route on the real board.



WIRE	FROM / HOLE	TO / HOLE
W001	U1.38 C03-R14	U1.3 C03-R21
W002	U1.3 C03-R21	U1.8 C08-R21
W003	U1.8 C08-R21	U1.13 C13-R21
W004	U1.13 C13-R21	U1.18 C18-R21
W005	U1.13 C13-R21	TP5.1 C12-R26
W006	U1.18 C18-R21	U1.23 C18-R14
W007	U1.23 C18-R14	U1.28 C13-R14
W008	TP5.1 C12-R26	J5.3 C08-R32
W009	U1.28 C13-R14	J1.2 C10-R04
W010	U1.18 C18-R21	J6.8 C31-R33
W011	J6.8 C31-R33	J4.12 C54-R31
W012	J4.12 C54-R31	J4.10 C54-R30
W013	J4.10 C54-R30	J4.8 C54-R29
W014	J4.8 C54-R29	J4.6 C54-R28
W015	J4.6 C54-R28	J4.4 C54-R27
W016	J4.4 C54-R27	J4.2 C54-R26
W017	J4.2 C54-R26	J3.12 C54-R20
W018	J3.12 C54-R20	J3.10 C54-R19
W019	J3.10 C54-R19	J3.8 C54-R18
W020	J3.8 C54-R18	J3.6 C54-R17
W021	J3.6 C54-R17	J3.4 C54-R16
W022	J3.4 C54-R16	J3.2 C54-R15
W023	J3.2 C54-R15	J2.12 C54-R09
W024	J2.12 C54-R09	J2.10 C54-R08
W025	J2.10 C54-R08	J2.8 C54-R07
W026	J2.8 C54-R07	J2.6 C54-R06
W027	J2.6 C54-R06	J2.4 C54-R05
W028	J2.4 C54-R05	J2.2 C54-R04
W036	J1.1 C12-R04	D2.2 C20-R08
W037	U1.39 C02-R14	D2.1 C16-R08
W038	D2.1 C16-R08	TP1.1 C24-R06
W061	U1.40 C01-R14	TP2.1 C24-R10

Each row is one separate wire. No bare-wire crossing or contact with an unused solder pad.

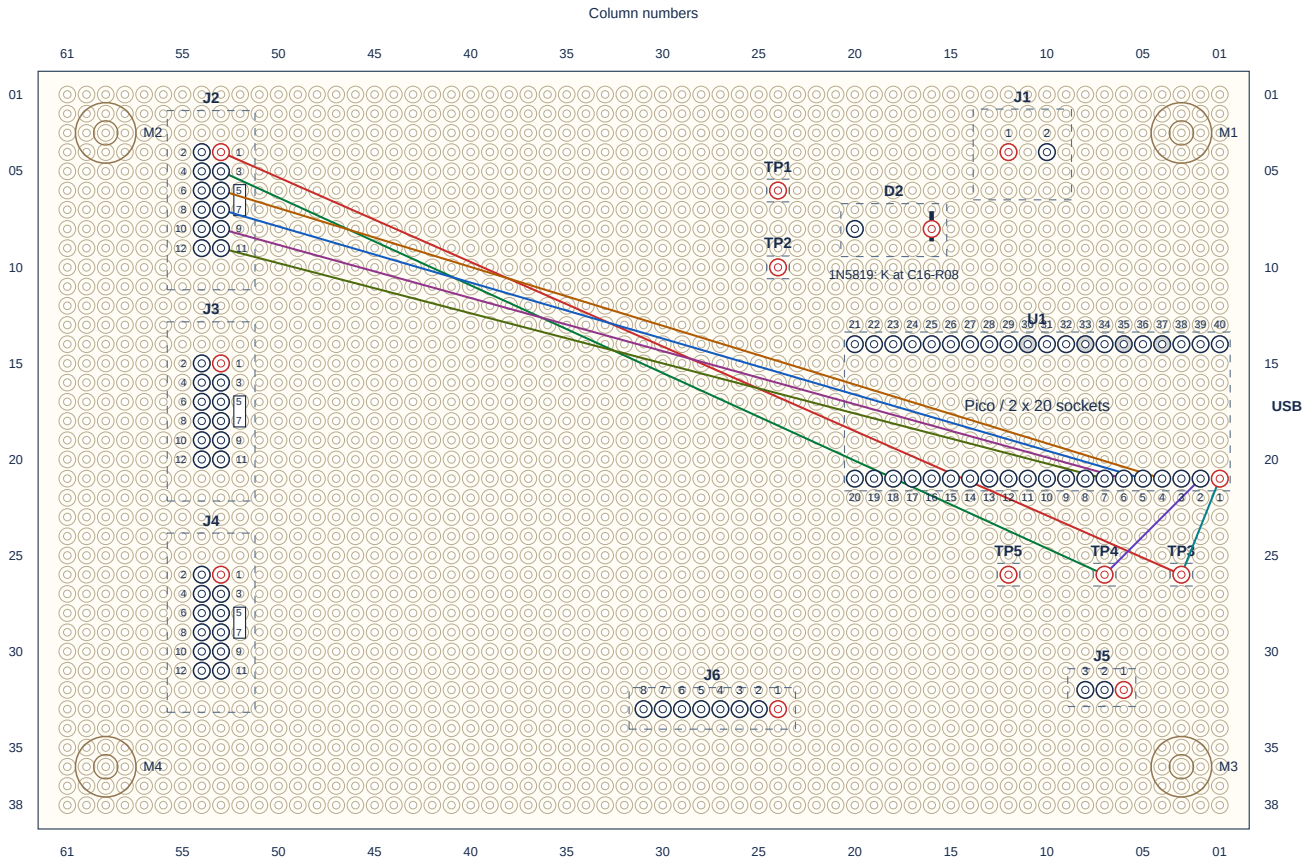
Hover a wire in the SVG for its ID/net. wiring.csv is the complete authoritative endpoint list.

Print at 100% / Actual size. Disable Fit to page. Check the scale and three distant holes before use.

50 mm scale check

Solder-side wiring: Port A / GP0..GP5

Mirrored view. Lines show endpoint connections; choose a practical insulated-wire route on the real board.



WIRE	FROM / HOLE	TO / HOLE
W039	U1.1 C01-R21	TP3.1 C03-R26
W040	TP3.1 C03-R26	J2.1 C53-R04
W049	U1.2 C02-R21	TP4.1 C07-R26
W050	TP4.1 C07-R26	J2.3 C53-R05
W051	U1.4 C04-R21	J2.5 C53-R06
W052	U1.5 C05-R21	J2.7 C53-R07
W053	U1.6 C06-R21	J2.9 C53-R08
W054	U1.7 C07-R21	J2.11 C53-R09

Each row is one separate wire. No bare-wire crossing or contact with an unused solder pad.

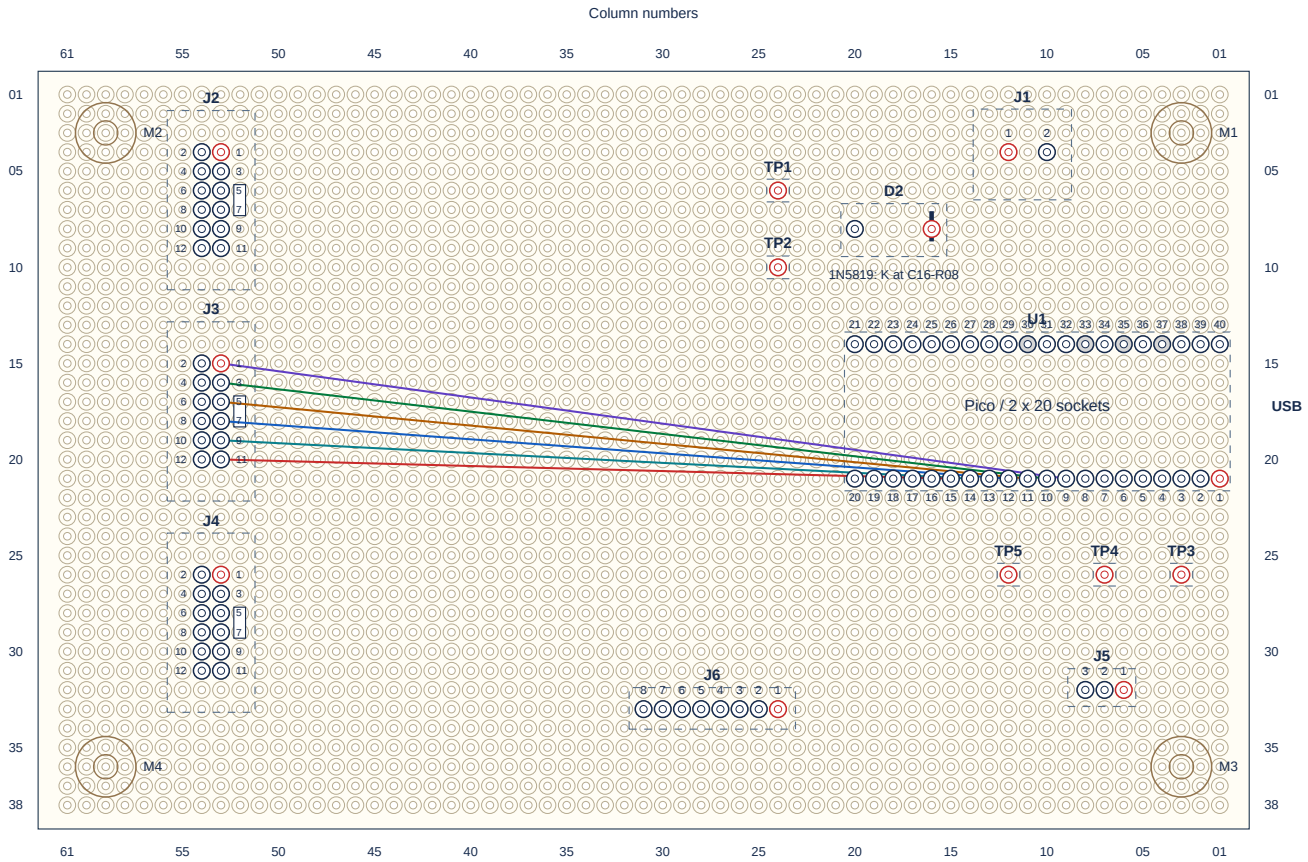
Hover a wire in the SVG for its ID/net. wiring.csv is the complete authoritative endpoint list.

Print at 100% / Actual size. Disable Fit to page. Check the scale and three distant holes before use.

50 mm scale check

Solder-side wiring: Port B / GP6..GP11

Mirrored view. Lines show endpoint connections; choose a practical insulated-wire route on the real board.



WIRE	FROM / HOLE	TO / HOLE
W041	U1.14 C14-R21	J3.9 C53-R19
W042	U1.15 C15-R21	J3.11 C53-R20
W055	U1.9 C09-R21	J3.1 C53-R15
W056	U1.10 C10-R21	J3.3 C53-R16
W057	U1.11 C11-R21	J3.5 C53-R17
W058	U1.12 C12-R21	J3.7 C53-R18

Each row is one separate wire. No bare-wire crossing or contact with an unused solder pad.

Hover a wire in the SVG for its ID/net. wiring.csv is the complete authoritative endpoint list.

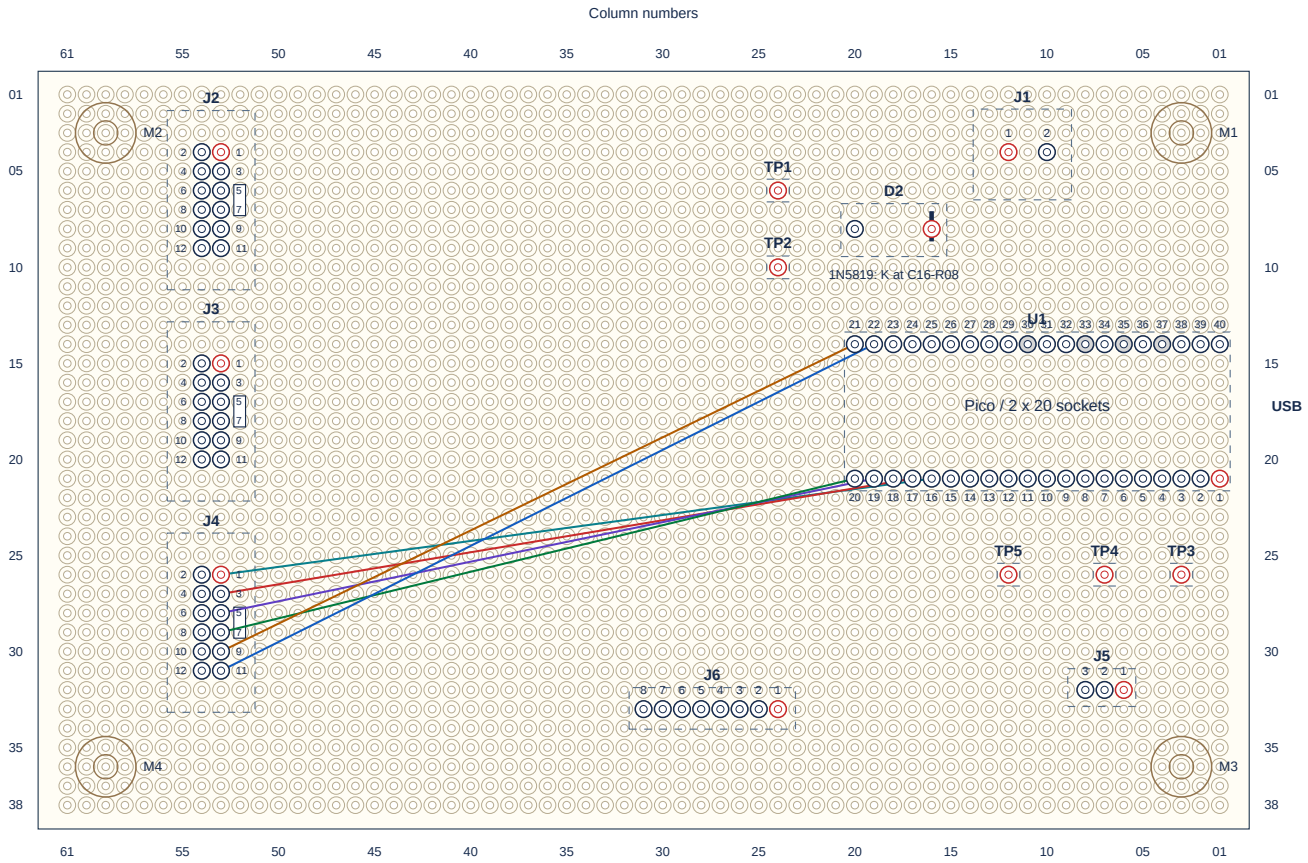
Print at 100% / Actual size. Disable Fit to page. Check the scale and three distant holes before use.

50 mm scale check

Solder-side wiring: Port C / GP12..GP17

Mirrored view. Lines show endpoint connections; choose a practical insulated-wire route on the real board.

P1 - UNBUILT / 2026-10-27



WIRE	FROM / HOLE	TO / HOLE
W043	U1.16 C16-R21	J4.1 C53-R26
W044	U1.17 C17-R21	J4.3 C53-R27
W045	U1.19 C19-R21	J4.5 C53-R28
W046	U1.20 C20-R21	J4.7 C53-R29
W047	U1.21 C20-R14	J4.9 C53-R30
W048	U1.22 C19-R14	J4.11 C53-R31

Each row is one separate wire. No bare-wire crossing or contact with an unused solder pad.

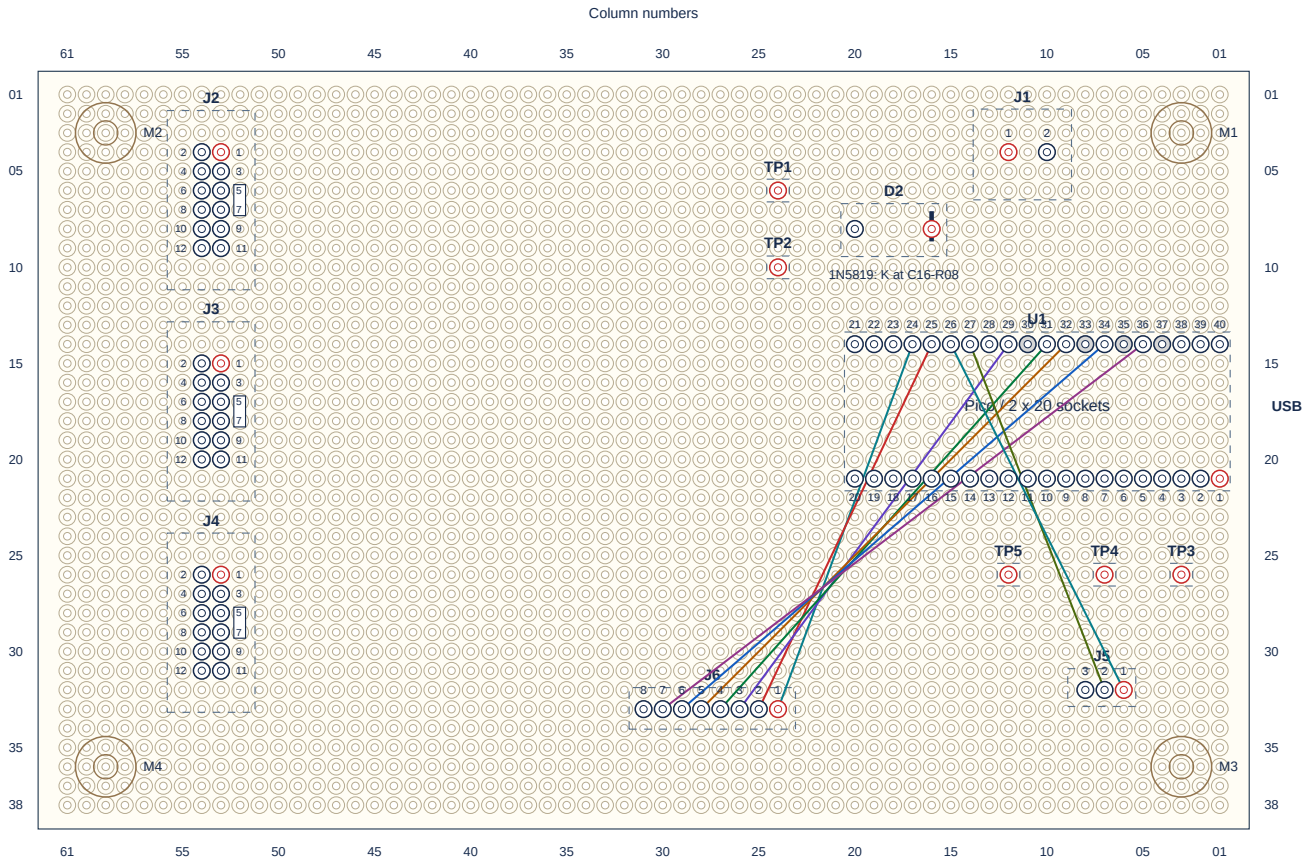
Hover a wire in the SVG for its ID/net. wiring.csv is the complete authoritative endpoint list.

Print at 100% / Actual size. Disable Fit to page. Check the scale and three distant holes before use.

50 mm scale check

Solder-side wiring: UART and expansion

Mirrored view. Lines show endpoint connections; choose a practical insulated-wire route on the real board.



WIRE	FROM / HOLE	TO / HOLE
W029	U1.24 C17-R14	J6.1 C24-R33
W030	U1.25 C16-R14	J6.2 C25-R33
W031	U1.29 C12-R14	J6.3 C26-R33
W032	U1.31 C10-R14	J6.4 C27-R33
W033	U1.32 C09-R14	J6.5 C28-R33
W034	U1.34 C07-R14	J6.6 C29-R33
W035	U1.36 C05-R14	J6.7 C30-R33
W059	J5.2 C07-R32	U1.27 C14-R14
W060	J5.1 C06-R32	U1.26 C15-R14

Each row is one separate wire. No bare-wire crossing or contact with an unused solder pad.

Hover a wire in the SVG for its ID/net. wiring.csv is the complete authoritative endpoint list.

Print at 100% / Actual size. Disable Fit to page. Check the scale and three distant holes before use.

50 mm scale check